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C.S.R. Prabhu

E-Governance

CONCEPTS AND CASE STUDIES



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E-GOVERNANCE: Concepts and Case Studies
C.S.R. PRABHU

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insight into everything connected with e-governance. His book, *E-Governance: Concepts and Case Studies* is, therefore, an excellent addition to the literature on the subject. To say that it is one among many is an understatement. It is certainly one of the best on the subject.

The Government of Andhra Pradesh has taken to e-governance as part of its vision 2020 for the all-round development of the State. The initial contribution for this has come from the NIC, especially from Mr. Prabhu and his colleagues. The assignment of the e-governance projects to private companies later, in addition to the NIC, has also resulted in competition for excellence to the benefit of NIC as well as to the government.

The book includes theoretical concepts and description of practical experience gained in designing and implementing e-governance projects in several Departments of the Union and state governments. Of particular value are the case studies which cover a broad spectrum of e-governance implementation, including projects in countries outside India. These add special value to the book.

Today, all engineering colleges have courses on computers, software and programming. This book will greatly help students and teachers to use their knowledge in a better fashion. The book brings in a touch of realism and practice to what is being taught in the colleges. The author, who is passionately devoted—in fact, is ‘paranoid’ about e-governance (recall what Andy Grove of Intel Corporation has said “*only the paranoid will succeed*”) deserves to be congratulated for writing this excellent book. He should be a model for many of the very good leaders in the design and implementation of e-governance projects and for writing a book that adds substantially to the knowledge on the subject. Indeed, I have no hesitation in saying that Mr. Prabhu has done a commendable job.

Dr. T.H. Chowdary

Director, Centre for Telecom Management and Studies, and
Chairman Pragna Bharati, Andhra Pradesh

Preface

E-governance is the latest trend in the governance process all over the world. Good governance can be enabled by e-governance if appropriately implemented. Good governance will be SMART (Simple, Moral, Accountable, Responsive and Transparent) governance which is so essential today in countries all over the world.

All the developed nations of the world such as USA, the UK, Canada, Australia and Singapore have gone in a big way into e-governance. Also, developing nations like India, China, Sri Lanka, the Philippines, Brazil have also progressed well in e-governance implementations.

It is in this context that the need for a book on e-governance becomes obvious, and this book is an attempt to fulfil such a need. I have tried to give a clear picture of e-governance, with the exposition of the principles and the provision of a number of case studies on e-governance. The discussions in the book range from the definition of e-governance to its history, evolution, various e-governance models, infrastructure and manpower facilities.

New technology domains such as data warehousing and data mining are also presented from the perspective of their applicability to e-governance domains with a possible plan of action in this direction. Besides, the factors that influence the success or failure of e-governance projects are dealt with.

One of the key features of this book is the provision of 22 case studies—18 Indian case studies and four international case studies. The Indian case studies include *Bhoomi*, a project of Karnataka Government, CARD (Computer-Aided Administration of Registration Department), *Smart Nagarpalika* (Computerization of Urban Local Bodies or Municipalities), IT in Judiciary, *Sachivalaya Vahini* (e-governance at Secretariat), *e-Khazana* (computerization of Treasury Department), and *e-Panchayat* (Electronic Knowledge-Based Panchayat). The international case studies are taken from USA, China, Brazil and Sri Lanka.

This book is intended for students of computer science, IT courses, management and public administration. In addition, government departments—both at the Centre and in various states—and administrators should find the book highly useful.

It is my fervent hope that this book with its emphasis on both principles and case studies would greatly benefit the readers.

Any constructive suggestions for the improvement of the book will be warmly appreciated.

C.S.R. PRABHU

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My special thanks are due for D.C. Misra of NIC and Prasad of *e-Commerce* magazine for permitting the survey of their work on modelling. Finally, I wish to thank Dr. Clay G. Wescott for granting permission to use his work.

C.S.R. PRABHU

CHAPTER 1

What is E-Governance?

1.1 E-GOVERNANCE—AN OVERVIEW

E-governance is a form of e-business¹ in governance comprising of processes and structures involved in deliverance of electronic services to the public, viz. citizens. It also involves collaborating with business partners of the government by conducting electronic transactions with them. Besides, it entails enabling the general public to interact with the government, through electronic means, for getting the desired services. In other words, e-governance² means application of electronic means in the interaction between

1. government (G) and citizens (C), both ways (i.e. G2C and C2G),
2. government or businesses (B), both ways (i.e. G2B and B2G), and
3. internal government operation (G2G),

The aim, ultimately, is to simplify and improve governance and enable people's participation in governance through mail, and Internet.

E-governance is much more than just preparing some websites. It ranges from the use of Internet for the dissemination of plain web based information at its simplest level to services and online transactions on the one hand and utilizing IT in the democratic process itself, i.e. election on the other.

E-governance implies e-democracy³ (Backers 2001), wherein all forms of interaction between the electorate (i.e. general public) and the elected (i.e. the government) are performed electronically. E-government, as distinguished from e-governance, comprises a pragmatic application and usage of the most innovative technologies in computer and communication technologies, including Internet technology, for delivering efficient and cost effective services, and information and knowledge to the citizens being governed, thereby realizing the vast potential of the government to serve the citizens.

Various manifestations of e-governance initiative will be in terms of the government delivering services to citizens of transacting business, offering general information, or conducting interactions with the general public and business using such IT tools as:

- E-mail
- Internet web sites publishing (including online interactive transaction)
- WAP application and publishing
- SMS connectivity
- Intranet development and usage
- Promotion of citizen access.

The advent of these other components and of Information and Communication Technology (ICT) as a highly leveraged enabling tool for delivery of services in the public and private sector has now been universally recognized. This has resulted in a redefinition of the fundamental concept of governance and also in recognizing its potential to change both institutions and delivery mechanisms of services for betterment of people.

1.1.1 Why E-governance?

The fundamental motivation for the campaign of e-governance in India and elsewhere is a slogan—to provide **SMART government**—“SMART” being an acronym for Simple, Moral, Accountable and Responsive Government, a laudable ideal, though difficult it may be to achieve in reality. Thus we may conceive a Smart Village or Smart Municipality or a Smart State, all very difficult, but ideal models. Notwithstanding the difficulties involved in achieving this, a clear objective of e-governance can be cutting the cost of governance and also minimizing the complexities of procedures by possible business process reengineering. The concomitant benefit is empowerment of people through what is called ‘disintermediation’; in other words, eliminating the middleman or tout between the government and the people. For example, by doing so, property tax assessment and collection system can reduce the element of corruption in the system apart from increasing consumer convenience. The online system based on Internet will reduce contact with mediating officials, thereby reducing the possibility of malpractice. This does not however mean that the primary objective of e-governance is tackling corruption, even though it may be a fallout (though not necessarily).

Evidently, the objectives of achieving such e-governance go far beyond mere simple computerization of stand-alone back office operations in government offices. It should mean a drastic change in the way the government operates, and this means a new and redefined set of responsibilities for the executive, legislative and the judiciary. This requires bringing about a social catharsis, which needs to be done in a comprehensive, concerted and planned manner.

Historically, it was in Chile that a real e-governance initiative was taken up as early as in 1972, when the IT applications were unheard of in government and were limited even in business. They used techniques of IT not to just make government paperless or less of paper (as is presently being done) but to perform government work efficiently. They realized that transparency is the ability to regulate the conditions, not the transactions. Prof. Stafford Beer

implemented for President Allende of Chile, the first e-governance software that would help the government survive a severe crisis. The question that was asked to and answered by the software was whether the government would survive by getting adequate grip and control over the situation in time of a severe inflationary crisis due to economic blockade resulting from stopping of copper exports (which was accounting for 80% of the foreign exchange earnings of Chile). The software which was developed did help in restoring prices back to normal, thus making the government survive. Chile thus became the first country to have successfully implemented e-governance.

Even though the Chile experiment of the real e-governance early in 1972 was a success story, the subsequent efforts in implementing e-governance in various countries, including the developed ones, were not aimed at such profound or sweeping purposes of critical nature. Generally, the e-governance applications have been more mundane, simple and straightforward. As the winds of e-governance and e-government blow widely through public organizations across the world, more and more governments in different countries have been harnessing the Internet and the powers of IT to provide services of varied nature as follows:

- G-to-G (Govt. to Govt.—within and across the Govt.)
- G-to-C (Services by the Govt. to Citizens)
- C-to-G (Interaction of Citizens with the Govt.)
- G-to-B (Services of the Govt. to Business)
- B-to-G (Business interaction with the Govt.)

1.1.2 Issues in E-Governance Applications and the Digital Divide

Initially, the e-Governance activity starts with providing information services by the government departments to the public in terms of State websites. These websites provide information about the department concerned, its aims, objectives, citizens' charters, organizational details, facilities available and services provided to the public along with the fees payable, etc. However, as the role of IT in the specific organization increases, the web sites of government departments attempt at providing more advanced services such as dynamic information and also specific transactions such as making utility payments. Gradually, this e-interaction of the public with the government leads to organizational transformation, transparency of public services, speed of service performance, increased citizen participation in the government, and thereby greater facilitation of participative democracy. Ideally, as the public agencies such as government departments and public sector undertakings begin implementing e-governance and e-government initiatives, their performance improves and they are better equipped to interact with citizens and provide services over the Internet. Thus, the citizens are enabled access to government documents, file taxes, make payments as utility bills, obtain or renew licenses and permits of different kinds, make bookings and reservations for public

services, lodge complaints or file applications for various benefits, and even employment.

How much of these actually happen? What are the problems encountered in achieving them? The enthusiastic initiatives in e-governance and e-government are not without consequential problems, as any technological innovation has. These initiatives have the potential to create a digital divide⁴ within the society, especially in the poor and developing countries. While the e-governance initiatives may benefit certain privileged sections of the society, the underprivileged, those who do not have access to Internet or not well qualified or equipped to use Internet will be all the more distanced from the government, leading to disenchantment. Also, this will aggravate even further the existing divide between the privileged and the underprivileged. Thus, it is essential that governments concerned ensure that all citizens of different socio economic and educational strata will have adequate access to the basic skills and infrastructure to participate in an increasingly technological society. As the digital divide becomes perceptible in different countries, public policy makers need to devise policies that would address issues of universal access and educational needs of their citizens, so as to match the requirements of an IT enabled e-government and e-society.

In addition, as the e-governments make Internet as the primary access point for all citizens to interact with the government, the issues that need to be focussed are:

1. How will the performance of the government departments/public bodies be improved by e-governance initiatives?
2. What are the organizational effects of e-government and IT?
3. What are the correct strategies for success in e-governance projects?
4. What are the skills that are required by the government employees in an e-governance environment in the Information Age?

While detailed research is required to address these issues, preliminary indications are already available that e-governance increases efficiency, speed, effectiveness and citizen satisfaction. However, these will be true only if the e-governance services provided to citizens are fool-proof, reliable and inexpensive. The structural effects of e-governance and IT in government departments and public agencies are yet to be identified, and their long-term effects and longitudinal effects are yet to be studied. More research is required to be taken up to answer questions as to whether e-governance leads to decentralized decision making, and whether it results or calls for business process reengineering in the government departments and public organizations.^{5,6}

IT implementations do indicate the required business process reengineering (BPR) within the concerned government department. However, whether such reengineering is viable, can be implemented without any repercussions, what legal changes is required—these issues are open for discussion. Often, the prerequisites for reengineering of the processes in the government are not easy to meet: radical changes in the processes or procedures are not acceptable as they may lead to considerable repercussions, sometimes too radical to be acceptable or implementable, with many side effects.

A number of organizations are involved in studying these issues. The e-governance initiative, a part of National Centre for Public Productivity at Rutgers University, Newark, New Jersey; Centre for Digital Government, a US National Research and Advisory Institute; Centre for e-government, an international body; Centre for Electronic Governance, IIM-Ahmedabad; Centre for Good Governance and also National Institute of Smart Governance, both at Hyderabad, India, Centre for e-governance at Department of Information Technology, and Ministry of Communications and Information Technology, Govt. of India, New Delhi, are some such institutions.

1.2 EVOLUTION OF E-GOVERNANCE, ITS SCOPE AND CONTENT

Even though historically it was Chile which implemented real e-governance solution as early as the seventies, the current interest and attention on e-governance applications all over the world has its roots in the "Information Super Highway" concept initiated by the US Vice President Al Gore in early 1990s. The Information Super Highway was defined largely in terms of the information infrastructure at the national level by many countries including the US, UK, Canada, Australia and India. The focus was then largely on development of components of the infrastructure, such as fibre optic networks across the States or Nations. Subsequently, the interest was widened to include socio-economic considerations encapsulated in the concept of *Information Society* or *Knowledge Society*, which naturally has to encompass e-governance. That is how e-governance concept came into being in a formalized and focussed manner, even though attempts to implement Information Systems in the government departments and other public organizations have been made with partial success in various countries including India. Such earlier attempts did not receive the state patronage on a broad-based manner while individual or stray attempts may be cited to have succeeded.

In general, during 1980s and 1990s, the governments all over the world lagged behind the commercial world in accepting and implementing Information and Communication Technology (ICT). The commercial world, including the industrial world, had gone far ahead of the governments all over the world in harnessing the potential of ICT in their core and also peripheral activities. Commercial enterprises utilized ICT increasingly to reach out to their customers and business partners, thereby impressively enhancing their service quality, speed and convenience. E-commerce thus became a big boom (even though the boom never reached the expected levels). However, visible success cases of ICT application include the 24 hours ATM (Automated Teller Machines) services, 24 hours call centres, electronic shopping on the Web, the use of DTV, integrating cable TV with Internet, etc. The list could be unlimited. Examples can be cited for typical information systems that run the 'back offices' in the financial and other sectors of business and industry. In fact, such 'back office' computerization could be even handled offshore in developing countries like India, where the skilled software manpower and also unskilled

operational manpower have been available at low cost. The cost-effective satellite communication infrastructure facilitated such remote development and maintenance of software of these banking, financial, aviation and industrial sectors. This formed the bulk of the 'software exports' activity in countries such as India, Ireland, Israel, and China. Similarly, in 1990s and 2000 till now, the IT enabled services (ITES) formed the major component of remote services such as call centres, data entry, etc. However, as indicated earlier, all these activities were not concerned with e-governance. Governments were the last in the bandwagon of institutions attempting to harness ICT in their activities. However, though late, the governments all over the world finally woke up to realize the potential of ICT in all their activities.

The initial efforts of e-governance simply resulted in only partial automation of the existing paper based manual procedures and did not result in any significant reengineering or optimization. While implementation of ICT in the business has resulted in good amount of Business Process Reengineering (BPR) as to move away from redundant and inefficient functional business units and to restructure organizations around processes that support core business, in the government enterprises such radical or significant changes have failed to happen to a large extent. This situation could be traced to various factors in government functioning such as conservatism, resistance to change, and rigidity of legislation which impedes the amendment of rules and procedures.

As a result, ICT based management methodologies⁷ such as Business Process Reengineering, Supply Chain Management, Just In Time (JIT) methodologies, which had salutary effects in business and industry had left the government system practically untouched. The scope and extent of e-governance have been largely limited to simple applications with the maximum of computerized MIS and database management within the government departments along with gradually enhanced usage of simple ICT technologies such as e-mail, and limited usage of Internet and video-conferencing for government functions.

In addition to potentially delivering significant improvements in government services, ICT has been visualized by some as having much deeper and wider impact on society and even capable of affecting the quality of life and nature of democracy.

However, the significant issues that has become highly relevant for large scale implementation of ICT in governance are the issues of security, privacy, vulnerability of public ICT infrastructure to crime, potential for abuse, terrorism, and general crime, in addition to issues related to social cohesion, and social exclusion following what is popularly known as the *digital divide*.

Notwithstanding the issue of digital divide which basically refers to lack of access of poor people and rural people to Internet, the indirect benefits to all citizens from computerization and ICT in the government machinery will go a long way in improving the quality of life of people.

Thus, the scope of ICT implementation in government machinery can result in

- improvement of efficiency and effectiveness of the executive functions of government, including delivery of public services;
- greater transparency of government to citizens and business, permitting greater access to the information generated or collated by the government;
- fundamental changes and improvement in relations between citizen and the state thereby improving the democratic process; and
- better interactions and relationships amongst different
 - wings of the same government,
 - state or local governments within a country,
 - countries whose governments are web-enabled.

Any e-governance activity/project involves appropriate

- hardware and corresponding system software,
- networking of the hardware identified above—both the Internet and Intranet environment, and
- application software along with appropriate database management software.

1.3 PRESENT GLOBAL TRENDS OF GROWTH IN E-GOVERNANCE

Press reports (during the end of 2002) indicate a trend of global growth in e-governance utilization by people in different categories. They indicate the following: The proportion of adults worldwide using the Internet to access government services or products during the past 12 months has increased by around 15 per cent, according to the findings of the second Government Online Study published by Taylor Nelson Sofres. Three out of ten citizens (30 per cent) globally said that they had accessed government services online compared with only a quarter (26 per cent) questioned a year ago.

Government online services are most commonly used to search for information (24 per cent of users) and to download information (11 per cent of users). The increased use of government online services is primarily due to the rise in the proportion of people searching for information (from 20 to 24 per cent during the period from September 2001 to September 2002). Globally, online government transactions increased from just 6 per cent to 7 per cent during this period and the percentage of those providing personal details to government increased from 7 per cent to 8 per cent.

In some countries, percentage increase has been significantly higher than in others. Among the most significant increases in the use of government services online are Australia (from 31 per cent to 46 per cent), Turkey (from 3 per cent to 13 per cent), the Netherlands (from 31 per cent to 41 per cent), and the US (from 34 per cent to 43 per cent). In contrast, in Japan, however, government online usage fell by 4 per cent (from 17 per cent to 13 per cent of citizens) between 2001 and 2002.

While security issues about accessing government services online were

the main concern for many countries during 2001, perceptions of safety improved globally during 2002. When 23 per cent of citizens worldwide said that they feel safe disclosing personal information such as credit card and bank account numbers online compared to just 14 per cent of citizens in 2001, representing thus an increase of almost two-thirds (64 per cent). As for the use of government online, the Scandinavian markets (Denmark, Finland, Norway, and Sweden), together with some South East Asian markets (Singapore and Hong Kong), have perceived the highest levels of safety (around one-third of users), in the system. In contrast, the greatest safety concerns were expressed by citizens in Japan (90 per cent said they felt accessing government service online was "unsafe"), Germany (82 per cent) and France (76 per cent).

1.3.1 Other Key Findings

Globally, government online use is more prevalent among men (33 per cent) than women (26 per cent), and among those aged under 35 compared with other age groups.

During the past 12 months substantial increases in government online use have taken place among 35–44 year-olds (from 22 per cent to 36 per cent) and 55–64 year-olds (from 2 per cent to 18 per cent). In contrast, use among those aged 65 and above decreased (from 7 per cent in 2001 to 5 per cent in 2002).

Globally, the proportion of Internet users who have made transactions using government services online is equal to the proportion of users who made online shopping transactions. Fifteen per cent of internet users have made an online government transaction and in addition 15 per cent have made an online purchase at least once during the past 12 months.

The percentage of Internet users who access government online services varies considerably across different countries from 16 per cent in Hungary to 81 per cent in Norway.

Wendy Mellor, Director, Taylor Nelson Sofres commented: The increase in the use of government online services at a global level suggests that the public see the Internet as a more acceptable means of getting involved in government activity at both national and local levels. However, significant differences exist between countries, may be due to, awareness of services, perceptions of safety, relevance of the site to users, and access to the Internet, among others.

In countries such as Singapore, Norway and Sweden, where the use of government services online is high, it is likely that a significant proportion of citizens feel comfortable with this approach of dealing with government. Yet in countries such as Britain, New Zealand and South Korea, where usage lags behind general Internet use, more needs to be done to assess why uptake of online services is slow and what steps need to be taken to address this.

All the above statistics on usage is time bound. Over the years there has been a definite rise in the usage of e-governance all over the world.

Conclusion

While the growth in the use of e-government is encouraging, our research shows that the majority of this growth is from citizens searching for information online rather than making transactions or providing personal information to government. This may be due to perceived security risks but if the use of these services is to increase, messages about the safety of government online services need to be communicated effectively.

CHAPTER 2

E-Governance Models

2.1 INTRODUCTION

In Chapter 1 we have defined e-governance fundamentally as application of ICT to governance activity. However, this can be manifested in multifarious ways and models. Models for e-governance, especially in the developing countries, are essential for a right perspective on e-governance implementation. In this chapter we shall survey some models for e-governance in developing countries.

2.2 MODELS OF DIGITAL GOVERNANCE

Models of digital governance⁸ are still evolving in developing countries. A few generic models have shaped up, which are finding greater recognition and are being replicated. These models are based on the inherent characteristics of ICT such as enabling equal access to information to anyone who is a part of the digital network and de-concentration of information across the entire digital network, connecting all sources of information. In simpler terms, information does not reside at any one particular node in the Digital Governance Models but flows equally across all the nodes—a fundamental change from the more common hierarchical information flow model that leads to unequal distribution of information and hence skewed power relations.

Hierarchy is inherent in the government departments. Equity based information flow may not be always compatible with government functioning. Therefore, appropriate administrative reforms and some reengineering may be required before e-governance may be really implemented.

It needs to be noted here that these models of governance are fundamentally different from those which are popular in developed countries due to differences in basic conditions, and perspectives and expectations from good governance. The six generic models of digital governance in developing countries are:

1. Broadcasting/Wider Dissemination Model

2. Critical Flow Model
3. Comparative Analysis Model
4. Mobilisation and Lobbying Model
5. Interactive-Service Model
6. E-governance Maturity Model

These models exhibit several variations dependent on the local situation and the governance functions carried out through these models.

2.2.1 Broadcasting/Wider Dissemination Model

Principle

The model is based on dissemination of information relevant to better governance that is already in the public domain into wider public domain through the use of ICT and convergent media. The rationale behind the model is that a more informed citizenry is able to understand better the governance mechanisms and is more empowered to make informed choices and exercise its rights and responsibilities. Further, there is a greater likelihood that the society in which the individuals are equally informed will ensure that the agenda and forms of governance are not biased to favour a few.

The wider dissemination model opens up an alternative channel for people to access information as well as validate information available in the local domain from external sources. The widespread application of this model gradually corrects the situation of information failure and provides people with the basic government-related information to come to a common understanding and decide upon the future course of action.

Applications

1. Putting government laws and legislation online.
2. Making available the names, contact addresses, e-mails, and fax numbers of local governmental officials online.
3. Making available key information pertaining to governmental plans, budgets, expenditures, and performances online.
4. Putting key court judgements/judicial statements that are of value to common citizens and creating a precedence for future actions online, viz. key environment related judgements, State vs Citizen court rulings, etc.

Project GISTNIC⁹ (General Information Services Terminal of National Informatics Centre) is an example of this model. In this project, the government agency (NIC) disseminates general information of about 25 subjects such as Economy, Education, Census, Tourism, etc. to general public. Government Orders (GOs) also are being publicised. However, after the advent and popularity of Internet, almost all government departments have been setting up or maintaining websites providing information about themselves to the public in general. The web sites of government departments can be reached through www.nic.in, a general government web site.

Evaluation

This model is the first step to more evolved forms of digital governance models. It is also the most crucial one as it catalyses free access and flow of information to all segments of society and serves as the building block to better governance.

The model, however, loses its effectiveness where free-flow of information is not encouraged or is not an objective. Tight governmental controls and bids to censor the content being transmitted through this model would be the bane of the model. The onus is therefore both on governmental organizations as well as civil society organizations to ensure that such models continue to proliferate.

2.2.2 Critical Flow Model

Principle

The model is based on channelling information of critical value to a targeted audience or spreading it in the wider public domain through the use of ICT and convergent media. The model requires foresight to understand the significance of a particular information set and use it strategically. It may also involve locating users to whom the availability of a particular information set would make a critical difference in initiating good governance.

The strength of Critical Flow Model is the inherent characteristic of ICT that makes the notion of distance and time redundant. This reduces the cases of exploitative governance possible earlier due to time lag between availability of information to different users.

Applications

The applications involve making available

- (a) information on corruption (by an appropriate legal authority) of a particular government ministry or government officials, to its electoral constituency or to the concerned governing body (e.g., the web sites of Central Vigilance Commission);
- (b) research studies, enquiry reports and appraisals commissioned by the government to the affected parties;
- (c) human rights violation and criminal impeachment records against government officials to NGOs and concerned citizens; and
- (d) environment related information to local communities, for example, information on radioactivity spills, effluent discharge in rivers, green ratings of a company, etc.

Evaluation

Critical Flow Model is more focussed in terms of its information content and its intended users. Due to critical aspect of information, the model exposes the weakest aspects of governance and decision-making mechanisms and informs

people about specific cases of state failure and bad governance to build up a case for concerted action. At the same time, by fuelling public unrest, the model exerts pressure on the concerned government institutions and individuals to take into cognizance the interest and opinion of the masses in decision making processes. The onus of creating such models may lie more with the civil society organizations to emerge as an effective watch guard to government policies and actions. The model will not work in cases where government mechanisms do not foster public debates and censure all information of critical nature. It will also fail where the government maintains a tight control over all information. There it remains restricted to top few levels of the government. Inherently the Internet is an open medium. Thus, restricted dissemination is only typical—only those interested may use the critical and subject based information lodged on Internet web sites for public access (as indicated above) as applications.

2.2.3 Comparative Analysis Model

Principle

The Comparative Analysis Model is based on exploring information available in the public or private domain and comparing it with the actual known information sets to derive strategic learnings and arguments. The model continuously assimilate new knowledge products and uses them as a benchmark to evaluate, influence or advocate changes in current governance policies and actions. The comparison could be made over a time scale to get a snapshot of the past and present situation (before-after analysis) or between two different situations to understand the effectiveness of an intervention (with or without analysis). The strength of this model lies in the boundless capacity of ICT to store information in a retrievable manner and transmit it almost instantaneously across all geographical and hierarchical barriers.

Applications

1. Gauging the effectiveness of current policies by gleaning learnings from government policies and actions of the past.
2. Establishing conditions of prior precedence, especially in the case of judicial or legal decision-making and use it to influence future decision-making. This could be useful in resolving patent-related disputes, public goods ownership rights, etc.
3. Enabling informed decision-making at all levels by enhancing the background knowledge and provide a rationale for future course of action.
4. Evaluating the performance record of a particular government official or ministry.

Evaluation

Developing countries can effectively use this model to their advantage as ICT

opens access to global and local knowledge products at a relatively low cost. Watchguard organizations and monitor groups could use the model to track the performance records of electoral candidates and share them in their constituency. The model is, however, dependent on the availability of comparative information sets and the ability of the users to analyze and bring out strong arguments or self-explanatory graphics from the analysis. The model however becomes ineffective in the absence of a strong civil society interest and short public memory.

2.2.4 Mobilization and Lobbying Model

Principle

Mobilization and Lobbying Model is one of the most frequently used digital governance models and has often come to the aid of civil society organizations in developing countries to impact international decision-making processes. The model is based on planned, directed, strategic flow of information to build strong virtual allies to strengthen action in the real world. It takes up the proactive approach of forming virtual communities which share similar values and concerns, promoting active sharing of information between these communities, and linking them with real-world activities.

The strength of this model is in the diversity of its virtual community, and the ideas, expertise and resources accumulated through virtual forms of networking. The model is able to effectively overcome geographical, institutional and bureaucratic barriers to shape concerted action. It also provides a strong virtual arm to several activities such as directing campaigns against a particular individual or decision-making body.

Applications

1. Fostering public debates on global issues, themes of upcoming conferences, treaties, etc.
2. Formation of pressure groups to pressurize decision-makers to take their common concerns into cognizance.
3. Amplifying the voices of marginalized groups such as backward classes or minorities who are traditionally marginalized from the decision-making process.
4. Encouraging wider participation in decision-making processes.
5. Developing global expertise on a particular theme in the absence of localized information to aid decision-making.

Evaluation

The Mobilization and Lobbying Model enhances the scope of participation of individuals and communities in policy issues and debates. The model also creates an effective deterrent for government bodies and individuals to be watchful in their actions lest they turn the opinion of local and global

community against them. This model could be effectively used by the Government to encourage public debates and to gauge public opinion on a particular issue as a part of good governance strategies.

2.2.5 Interactive-Service Model/Government-to-Citizen-to-Government Model (G2C2G)

Principle

Interactive-Service Model in many ways is a consolidation of the earlier digital governance models and opens up avenues for direct participation of individuals in the governance processes. This model fully captures the potential of ICT and leverages it for greater participation, efficiency and transparency in the functioning of government as well as savings in time and costs relating to decision-making.

The Interactive-Service Model makes possible various services offered by the government to be directly accessible to citizens. It creates an interactive Government-to-Consumer-to-Government (G2C2G) channel in various functions such as election of government officials (e-ballots), filing of tax returns, procurement of government services, sharing of concerns and providing expertise, conducting opinion polls on public issues, and grievance redressal.

Applications

1. Establishing an interactive communication channel with policy-makers such as videoconferencing and online dialoguing.
2. Conducting electronic ballots for the election of government officials and other office bearers.
3. Conducting public debates/opinion polls on issues of wider concern before formulation of policies and legislative frameworks.
4. Filing of grievance petitions, feedback and reports by citizens with the concerned governmental body.
5. Performing governance functions online such as revenue collection, filing of taxes, governmental procurement, payment transfers, etc.
6. Carrying out videoconferencing, and online discussion with policy makers.

Evaluation

This model is more embedded in e-governance initiatives in the developed countries and has often been proposed for implementation in developing countries. Such forms of solution transfer may not be very effective. The model is on the higher end of technology reliance as compared to the other models. This makes it difficult to replicate in developing countries in the absence of individual and secure ICT access. Various other issues also need to be considered carefully before such blind duplication can be attempted in the developing countries. However, the trend is definitely in this direction and

sooner or later, this model will be implemented in all the countries with due modifications for local adaptation.

2.3 EVOLUTION IN E-GOVERNANCE AND MATURITY MODELS

The E-governance Maturity Model (EMM—version 1.0)¹⁰ (D.C. Misra and Anjali Dhingra), based on the conventional software maturity models, proposes some levels of maturity, depending on the effectiveness with which the e-governance efforts have been initiated, implemented or successfully completed. The model also provides for identification of key focus areas that need to be concentrated for attaining a specific maturity level as discussed below:

Traditionally, e-governance has been defined as an ICT enabled route to good governance with a view to enhancing transparency in the system and provide prompt and quality services to the citizen. E-governance is an evolutionary path, whose effective implementation requires a complete understanding of its various elements and at the same time taking a holistic view to stay focused on its overall objectives.

E-governance journey encounters several milestones that need to be identified and modelled so that efforts invested can be assessed and an appropriate course of action be taken by the organization to further its way on the path of e-governance. The E-governance Maturity Model (EMM—version 1.0) proposes five levels of maturity, depending upon the effectiveness with which the e-governance efforts have been initiated, pursued, utilized and institutionalized. EMM 1.0 will facilitate government organizations to assess the current level of e-governance initiatives and accordingly make efforts for the future. The model further identifies the characteristics exhibited by organizations at various levels of maturity that will facilitate correct assessment of the current status. The model also provides Key Focus Areas (KFAs) that need to be focused on to attain a particular maturity level.

2.3.1 Five Maturity Levels

The E-governance Maturity Model (EMM—version 1.0) is based on the fact that speed, openness and ubiquity are some of the major capabilities of ICTs, which can be leveraged for generating transparency, responsiveness and accountability in the system, on the one hand and empowering the common man by providing faster access to right information at the right time, on the other. It is also based on a service-oriented approach, where public administration is seen as a professional activity and efficient delivery of services to the internal and external users (customers) is emphasized as a key performance indicator of the government department. The internal customers/users of an organization are its employees and the external users are the citizens, businesses, other government and non-governmental bodies that the organization needs to satisfy in its e-governance endeavour. However, there

may exist organizations that are performing well even without ICT application to its functions. Even such organizations may gradually take up to ICT for the sake of other reasons and other benefits.

The maturity levels, described below, provide a necessary mechanism to benchmark the efforts invested by an organization in implementing e-governance and subsequently sustaining it to the satisfaction of its customers/users.

Level 1: Closed

This is the stage when an organization does not use ICT as a facilitator for good governance and has no plans to do so in the near future. This situation may arise due to lack of exposure to ICTs and associated benefits that again may depend upon a number of reasons; remoteness from the mainstream in terms of location is primary, and lack of resources and strategic thinking could be some of the other issues. As a result, the organization is 'closed' in terms of being connected and sharing of information in the context of "e"-governance. However, even in this condition the organization may be efficiently functioning. Given the trend today, all organizations may take up ICT implementation sooner or later.

Level 2: Initial

This level corresponds to the stage when an organization has initiated the automation of its processes but on a ad-hoc basis. No organized efforts are made to undertake the e-governance initiatives. Also, due to lack of direction many such e-governance efforts are abandoned at a subsequent date.

Level 3: Planned

The e-governance initiative, at this level, is undertaken with a systematic approach. The organization has a clearly defined vision, objectives and goals for e-governance. A need assessment study is conducted to prioritize areas of implementation and gauge the extent of e-readiness. Taking input from the need assessment study, extensive planning has been carried out indicating policies, strategies, various activities, stakeholders, roles and responsibilities and resources required in terms of time, money and manpower to undertake the e-governance exercise. However, the organization is yet to enter into the planned implementation of the e-governance exercise, even though all the requisite planning is completed.

Level 4: Realized

This level corresponds to the stage when the organization actually realizes the complete e-governance plan. Consequently, an integrated system is established where all the internal processes of the organization are computerized and there is a seamless information exchange among all concerned entities. The organization starts delivering the services to its external as well as internal customers/users in an effective manner.

Complete realization of the plan, in a single instance, would entail enormous amount of resources in terms of time, money and manpower, which may necessitate adopting a phased approach for operationalizing the e-governance services. Accordingly, a further classification within this level has been proposed that measures the extent of realization of the plan over a period of time. These sub-levels are also indicative of the openness and effectiveness with which the information is exchanged among the various entities of the organization (external and internal).

- (a) **Retrospected.** At this level, the organization has retrospectively studied its business processes in view of its vision, overall e-governance objectives, the service-oriented approach (wherein government is expected to effectively deliver the services to its customers/users), and changes, if required, in the processes are initiated as a constant evolutionary process.
- (b) **E-ready.** In this stage, e-readiness essentials,* which are also the building blocks for e-governance, are ensured by the organization.
- (c) **Partially open.** At this stage some of the e-governance services are operationalized resulting in a partial information exchange among the entities, both within and outside the organization.
- (d) **Open.** This sub-level of realized state implies complete deployment of e-governance services that ensure an integrated system that is open to information exchange. The focus here shifts from acquiring and implementing “e” enabling factors to effectiveness with which the services are delivered. The system gains responsiveness to deal with the customer needs and is accountable for its services.

Level 5: Institutionalized

At this level, the organizations sustain the realized state over a period of time so that e-governance becomes part of its work culture. The e-governance services are effectively utilized and accepted by the users. Several iterations between planned and realized state lead to institutionalization, when e-governance becomes a way of life (Figure 2.1).

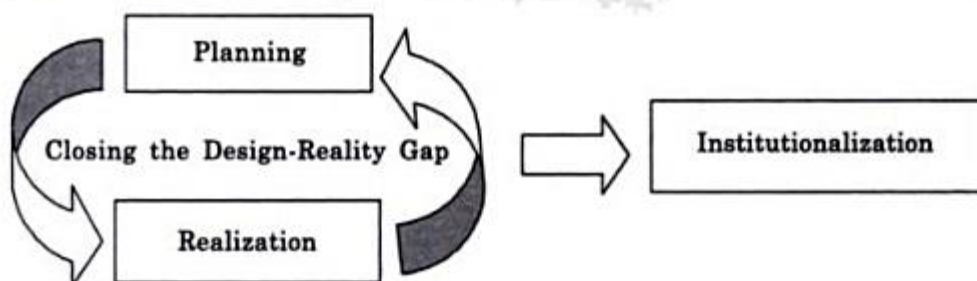


Figure 2.1 Reaching the Institutionalized Stage.

*E-readiness essentials are a set of prerequisites that act as sound building blocks for implementing e-governance. Most of the e-governance initiatives in the past have either failed or could not commence because of lack of e-readiness.

We have poor countries on the one hand and the middle-to-upper income countries on the other. The middle-to-upper level income countries are replacing manual services by electronic means. For example, in Hong Kong, 70 per cent of government services are already online, while Singapore has almost 100 per cent of the government services online through automatic kiosks and ATMs established in various public locations as local train stations. On the other hand, poor countries such as the ones described above are newly building linkages between people and the Government. Both the groups provide services through public kiosks.

Stage 5: Allowing online transactions by the citizens

In stage 4, the citizens interacted through kiosks, obtained services through online or the Internet, but made payments manually. But now, in Stage 5, in addition to permitting single online enquiry access to information, citizens may be enabled to make payments of fees and taxes, lodge complaints, file applications and perform any other transactions online through citizen kiosks installed at busy public locations. This is a much more advanced stage in e-governance not yet reached in developing countries but already reached in the developed world.

In Singapore, the citizens can transact every government business online and round the clock through specially designed kiosks which can be operated using smart cards. They can transact all government related business such as social welfare claims, tax assessment and payment, visa applications and license renewals, in addition to bank based fund or financial transactions using smart cards. However, this is a very advanced technological scenario, too advanced to be replicated immediately in all developing countries, even though plans are being drawn in developing countries to execute similar initiatives. Of course, limited transactions with the government by the citizens have been made online in these countries. The kiosks in public locations in Beijing, China, are successful; this is true of Korea as well as of India (for land record details).

Still, in these countries the financial transactions have not been made online, even though collections may be allowed alternatively (by an operator collecting money). The main reason for not executing financial transactions on the Internet has been the lack of security—lack of confident and secure financial transactions on the net. With the imminent implementation of public key infrastructure in the entire world, including the poor and developing countries, the confidence on financial transactions over the net may be developed, and with the proliferation of the smart cards, and online payments may be very soon a way of life in the entire developing world.

As regards the G2B (Government to Business) transactions, there have been impressive advances in various developing countries. For example, in the Philippines, the Customs Bureau has enabled electronic payments of customs duty, electronic processing of clearance of documents and release of shipments. The new online system has resulted in fast and secure transmission of payment details. The time for reconciling of payments collected by banks and remittances to the national treasury has been reduced to a few days from several

months. An information system called "Selectivity" categorizes shipments into high, medium or low risk transactions so that they can be coursed through appropriate examination procedures. This reduces fraud, corruption and other undesirable effects that normally result from the personal interaction of the officials with the business customers of the customs department.

Similar efforts, even though limited, have been made in other countries such as India, Thailand and Korea. In Thailand, the Customs Department has eliminated all manual processing, thus increasing efficiency and transparency. Similar effort is in place in Indian Customs Department which has started implementing Electronic Data Interchange (EDI). In Korea, the Public Procurement Service, a central government organization responsible for procuring commodities and arranging contracts for construction projects, has computerized the purchase process and accounting transactions using the EDI.

Cybershopping or e-procurement a growing trend in all countries, is popular for its ease of operation, efficiency and transparency. Contracting also is largely getting computerized. Databases of supply firms information are being set-up. Also, pre-qualification and cost accounting procedures are also being automated.

In countries such as Singapore, complete Enterprise Resource Planning (ERP) implementation has resulted in making compulsory the supplying firms to be e-enabled and compatible to the ERP environment (such as SAP).

Internet based tendering and contracting process can be designed in such a manner that the documents from contractors including performance records can be obtained using computer networks of relevant organizations directly instead of receiving them from middle men manually. This prevents false documents and data being supplied.

In many of these efforts, there is some scope for the government offloading some of its activities to private contractors who will operate the requisite services and earn their own revenues from the customers who may be individuals or business houses. There are models as Build, Own and Operate (BOO) or Build, Own, Operate and Transfer (BOOT) models. There have been some success stories in this regard all over the world. With increasing public sector reforms and privatization, these models may be more attractive than the conventional government owned operations of services. For example, in Hong Kong, the government web portal is entirely financed and maintained by a private company, thereby reducing the cost and risk to the government. Other governments also are expected to follow the lead and involve private sector in creating partnerships with suppliers and customers together with whom they can find ways to cut costs, improve quality and share benefits. The private partnership of the government should be taken up with caution as some of the information in government may be confidential and sensitive sometimes (though not always); also, the profit motive of private companies may not be always achieved by the government in developing countries. It is preferable that the government uses its own agencies for such purposes.

Stage 6: Enriching digital democracy

Democracy can be strengthened and enriched by ICT in multiple ways and modes. At least two important sets of ICT applications that can potentially support participatory and democratic processes, specially in the developing countries have been identified. These relate to applications that enrich and further empower the civil society organizations, and enable the citizens to express themselves by voting in democratic processes through Internet or by any other electronic means.

Examples of success stories on both these two applications identified above can be cited in developing countries. In the Grameena Bank Project in Bangladesh, a cell phone hand set could be bought by a poor woman (with the help of bank loan) who would in turn rent it out to other poor men and women who work in the fields. Finally, this resulted in a commercially viable rural cell phone service leading to significant economic development.

In Andhra Pradesh, the poor women in villages form self-help groups (called DWCRA Project) with micro credit. They were able to grow big enough to set up their own banks which use ICT to a significant level.

The application of ICT in civil society groups and organizations is of a wide spectrum. Several groups can even combine forces, raise funds and even challenge multinational corporations. These challengers can be as diverse as NGOs, trade unions and other self-help groups. Their networking of people could be achieved through ICT applications.

The Community Information Centre (CIC) Project in North Eastern States of India (set up by National Informatics Centre and Department of Information Technology, Ministry of Communications and Information Technology, Government of India) is a success story of engaging local youth in ICT application development and usage.

ICT has been used by the governments in various countries for strengthening democracy, democratic processes and reforms of democratic processes.

It is expected, in the long term, digital democracy will come to some countries in another form: electronic voting. As in USA and in Japan, voting sites can be set up for Internet based voting in elections. However, this will be very different in developing countries. In India and also in the Phillipines, the Election Commission has used ICT in several ways: the voters' information were kept in its web site along with details of timings and location of polling booths. Dissemination of such information through Internet was enhanced with the help of News Media. However, in developing countries largely the voting process itself is either manual or partly electronic. In India and other developing countries, even though Electronic Voting Machines (EVMs) have been used for voting purposes, the entire operation is still manual, even though Internet was used for purposes other than voting itself. Voting on the Internet has now made a beginning in limited scale in many small polls, especially in the developed countries.

Stage 7: Electronically integrated or joined up government with Legislature and Judiciary

In the final stage of e-governance a comprehensive web portal and a smart card integrates information and services from various responsible government agencies. In this stage both horizontal integration of services across departments and vertical integration of service delivery is expected to take place. As already available in Singapore and also being experimented in many states such as Andhra Pradesh, on a web portal users can obtain services across different geographical levels of government within the same functional area and also access different functions. Thus, in a scenario like this, a citizen could submit a change of address on driving license and such a change would automatically be effected in all other sectors such as health, education, elections, taxation, etc. (thereby avoiding need for multiple filing). This is true of horizontal integration of services in an e-government. Citizens also can use portals to make payments and other transactions, obtain a checklist of enclosures required for an application, find answers to frequently asked questions (faq) and engage the services of relevant commercial enterprises. In Singapore and Hong Kong, such state-of-the-art portals are operational. The Government of Taipei (Taiwan) also has set up a "One Window" service on the Internet for tax administration, public health and e-commerce. Smart cards are gradually becoming functional in several Asian Countries for all such activities.

In Andhra Pradesh the e-seva project offers single-roof service on about 40 different areas such as property registration, taxation, utility bills payment, etc. though not presently with a smart card and also not on a single portal, though attempts are in progress in these directions.

Finally, it can be pointed out that this highest level of e-government—Integrated Services with a smart card—is yet to become operational in most developing countries (Singapore and Hong Kong already have). Further integration of e-governance with e-judiciary and e-legislature is yet to come about.

CHAPTER 4

Applications of Data Warehousing and Data Mining in Government

4.1 INTRODUCTION

Data warehousing and data mining are the important means of preparing the government to face the challenges of the new millennium.

Data warehousing and data mining technologies have extensive potential applications in the government—in various Central Government sectors such as Agriculture, Rural Development, Health and Energy and also in State Government activities. These technologies can and should therefore be implemented.

In this chapter, we shall examine their potential applications in the State and Central Government.

4.2 NATIONAL DATA WAREHOUSES

A large number of national data warehouses can be identified from the existing data resources within the Central Government Ministries. Let us examine these potential subject areas on which data warehouses may be developed at present and also in future.

4.2.1 Census Data

The Registrar General and Census Commissioner of India decennially compiles information of all individuals, villages, population groups, etc. This information is wide ranging such as the individual-slip, a compilation of information of individual households, of which a database of 5% sample is maintained for analysis. A data warehouse can be built from this database upon which OLAP techniques can be applied. Data mining also can be performed for analysis and knowledge discovery.

A village-level database was originally developed by National Informatics Centre at Hyderabad under General Information Services Terminal of National Informatics Centre (GISTNIC) for the 1991 Census. This consists of two parts: primary census abstract and village amenities. Subsequently, a data warehouse was also developed for village amenities for Tamil Nadu. This enables multidimensional analysis of the village level data in such sectors as Education, Health and Infrastructure. The fact data pertains to the individual village data compiled under 1991 Census.

As the census compilation is performed once in ten years, the data is quasi-static and, therefore, no refreshing of the warehouse needs to be done on a periodic basis. Only the new data needs to be either appended to the data warehouse or alternatively a new data warehouse can be built.

There exist many other subject areas (e.g. migration tables) within the census purview which may be amenable and appropriate for data warehouse development, OLAP and data mining applications on which work can be taken up in future.

4.2.2 Prices of Essential Commodities

The Ministry of Food and Civil Supplies, Government of India, compiles daily data (on weekly basis) for about 300 observation centres in the entire country on the prices of essential commodities such as rice, edible oils, etc. This data is compiled at the district level by the respective State Government agencies and transmitted online to Delhi for aggregation and storage. A data warehouse can be built for this data, and OLAP techniques can be applied for its analysis. A data mining and forecasting technique can be applied for advance forecasting of the actual prices of these essential commodities. The forecasting model can be strengthened for more accurate forecasting by taking into account the external factors such as rainfall, growth rate of population and inflation.

A limited exercise in this direction was already executed at a State level (in Tamil Nadu).

4.3 OTHER AREAS FOR DATA WAREHOUSING AND DATA MINING

Other possible areas for data warehousing and data mining in Central Government sectors are discussed in detail as under.

4.3.1 Agriculture

The Agricultural Census performed by the Ministry of Agriculture, Government of India, compiles a large number of agricultural parameters at the national level. District-wise agricultural production, area and yield of crops is compiled; this can be built into a data warehouse for analysis, mining and forecasting. Statistics on consumption of fertilizers also can be turned into a data mart.

Data on agricultural inputs such as seeds and fertilizers can also be effectively analyzed in a data warehouse. Data from livestock census can be turned into a data warehouse. Land-use pattern statistics can also be analyzed in a warehousing environment. Other data such as watershed details and also agricultural credit data can be effectively used for analysis by applying the technologies of OLAP and data mining.

Thus there is substantial scope for application of data warehousing and data mining techniques in Agricultural sector.

4.3.2 Rural Development

Data on individuals below poverty line (BPL survey) can be built into a data warehouse. Drinking water census data (from Drinking Water Mission) can be effectively utilized by OLAP and data mining technologies. Monitoring and analysis of progress made on implementation of rural development programmes can also be made using OLAP and data mining techniques.

4.3.3 Health

Community needs assessment data, immunization data, data from national programmes on controlling blindness, leprosy, malaria can all be used for data warehousing implementation, OLAP and data mining applications.

4.3.4 Planning

At the Planning Commission, data warehouses can be built for state plan data on all sectors: labour, energy, education, trade and industry, five year plan, etc.

4.3.5 Education

The Sixth All India Educational Survey data has been converted into a data warehouse (with about 3 GB of data). Various types of analytical queries and reports can be answered.

4.3.6 Commerce and Trade

Data bank on trade (imports and exports) can be analyzed and converted into a data warehouse.* World price monitoring system can be made to perform better by using data warehousing and data mining technologies. Provisional estimates of import and export also be made more accurate using forecasting techniques.

*This data is available with the Director General of Foreign Trade, Ministry of Commerce.

4.3.7 Other Sectors

In addition to the above mentioned important applications, there exist a number of other potential application areas for data warehousing and data mining, as follows:

Tourism. Tourist arrival behaviour and preferences; tourism products data; foreign exchange earnings data; and Hotels, Travel and Transportation data.

Programme Implementation. Central projects data (for monitoring).

Revenue. Customs data, central excise data, and commercial taxes data (state government).

Economic affairs. Budget and expenditure data; and annual economic survey.

Audit and accounts. Government accounts data.

All government departments or organizations are deeply involved in generating and processing a large amount of data. Conventionally, the government departments have largely been satisfied with developing single management information systems (MIS), or in limited cases, a few databases which were used online for limited reporting purposes. Much of the analysis work was done manually by the Department of Statistics in the Central Government or in any State Government. The techniques used for analysis were conventional statistical techniques on largely batch-mode processing. Prior to the advent of data warehousing and data mining technologies nobody was aware of any better techniques for this activity. In fact, data warehousing and data mining technologies could lead to the most significant advancements in the government functioning, if properly applied and used in the government activities. With their advent and prominence, there is a paradigm shift which may finally result in improved governance and better planning by better utilization of data. Instead of the officials wasting their time in processing data, they can rely on data warehousing and data mining technologies for their day-to-day decision making and concentrate more on the practical implementation of the decisions so taken for better performance of developmental activities.

Further, even though various departments in the government (State or Central) are functionally interlinked, the data is presently generated, maintained and used independently in each department. This leads to poor (independent) decision making and isolated planning. Herein lies the importance of data warehousing technology. Different data marts for separate departments, if built, can be integrated into one data warehouse for the government. This is true for State Government and Central Government. Thus data warehouses can be built at Central level, State level and also at District level.

Conclusion

In the government, the individual data marts are required to be maintained by the individual departments (or public sector organizations) and a central data

- Single/Multi Session
- ISO/Joliet formats
- Speed 2X to Max Speed
- Test and Write/Write

Design of forms and reports

A set of bilingual (English and Telugu) forms and reports have been designed and printed in the required quantity.

The forms include

1. Requisition form for MVA.
2. Input form accompanying the document (Urban).
3. Input form accompany the document (Rural).
4. Indent for purchase of stamp paper.
5. Application for issue of EC (encumbrance certificate) and CC (certified copy).

Reports (Preprinted stationery) include

MV Report (Checkslip)
Registration Checkslip
EC
Cash Receipt.

A dedicated team of software professionals from NIC have developed and are constantly supporting the CARD software, training and human resource development.

C3.2.5 Human Resource Development

The most significant feature of CARD project is to develop adequate technical skills among the employees of the department for conducting day-to-day operations and maintenance of the CARD system.

The initial training programme was implemented by NIC, NIIT and STG in Hyderabad for five different levels of Officers/Staff of the department, as indicated in Table C3.3.

Table C3.3

<i>Category</i>	<i>Numbers</i>	<i>Training period</i>
Dy. Inspectors General and District Registrars	45	1 week
Asst. District Registrars	50	3 weeks
DPOs	270	8 weeks
Sub Registrars	389	2 weeks
DEOs	1200	2 weeks

This is further augmented by conducting workshops from time to time by NIC as and when new versions of CARD are released.

Maintenance of hardware and system software

- Up to 16-1-2000, the project was maintained by the multiple vendors who supplied hardware and system software.
- First facilities management came into force w.e.f. 17-1-2000 to 31-12-2001.
- Second facilities management contract came into force w.e.f. 1-1-2002 for a period of 2 years.
- Present facilities management for entire state is by WIPRO Infotech Ltd., Hyderabad.

C3.2.6 Launch of CARD Project

The CARD project was launched on 4th November 1998, defining the standards for different Registration Services with service levels as shown in Table C3.4.

Table C3.4

<i>Name of the Service</i>	<i>Time standard (minutes)</i>
Registration of documents	60
Issue of MVA	5
Issue of EC (In 222 SROs)	5
Sale of stamp papers	10
Issue of certified copy	10

All centres were inaugurated on the same day by public representatives of the area.

The legal effect was accorded to the CARD process of registration w.e.f. 5-2-1999.

The second phase of CARD covering 25 sub-registrar offices was launched on 1-11-2001.

Impact

The introduction of the CARD project using the Information Technology (IT) in one of the oldest departments of the government having employees with no technical skills, has served as a role model for other states in attracting senior officers of various states to visit CARD offices at Hyderabad.

Due to the keen interest shown by various states, NIC-APSU organized a presentation and demonstration of the CARD project in Kerala, Delhi, West Bengal, Orissa, and Goa.

Similar projects have been implemented in other states such as Tamil Nadu, Punjab, Haryana, Maharashtra, Orissa, etc. CARD project had been appreciated by many distinguished personalities of national and international level. The prominent among them are

- UN Secretary General

- President of USA
- President of India
- President of World Bank

More details about the CARD project and the department can be had from the official website www.ap.gov.in/card of the department.

It is a matter of pride for NIC to note that the CARD project has been mentioned as a model for e-governance in the World Bank web site with following URL:

www1.worldbank.org/publicsector/egov/cards.htm

C3.2.7 Road Map

- (a) Implementation of CARD in all remaining 148 SROs by 4-11-2002.
- (b) Networking of all CARD Sub-Registrar Offices to web-enable the following registration services (CARD Web Services):

- (i) Free Services

- Find SRO
- Land rates
- Duty rates for various transactions
- Structure rates

- (ii) Paid Services

Market value assistance
Encumbrance

- (c) Registration of documents on anywhere basis.

To support CARD Web Services and Registration on Anywhere basis the existing CARD software has been upgraded and implemented in all SROs by releasing i card 1.0 version. The feature "Anywhere Registration" is under trial run in the 11 SROs of district Hyderabad since July, 2002.

The software for providing CARD web services has been designed and made available for implementation on the web site <http://igrs.ap.gov.in> due for a formal inauguration by the government.

- (d) Development of MIS.
 - (e) Establishing E-mail connectivity.
 - (f) To establish central storage system for preservation of document images on permanent basis.

Limitations

As CARD system primarily deals with registration of immovable properties, it is essential to have a well-defined standard for describing the various types of properties. The absence of such standards will effect the process of valuation,

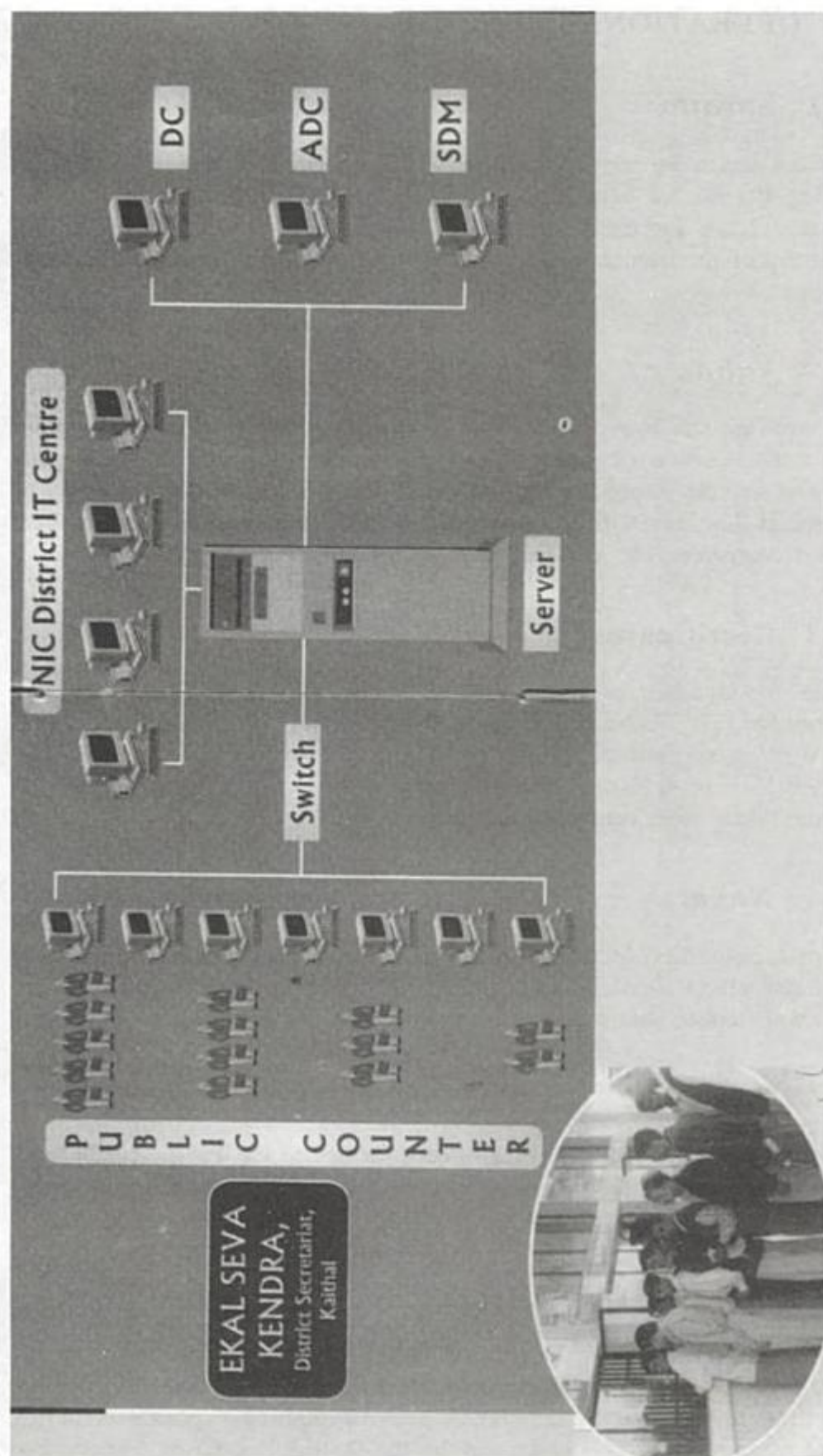


Figure C7.2 Technology behind Ekal Seva Kendra.

C7.3 OPERATIONAL ASPECTS

C7.3.1 *Sarathi*

A software, which provides comprehensive and integrated solutions for issuance of driving licence has been implemented. Driving test date is booked on the applicant's choice and the modularity of software has been designed to ensure the delivery of the licence in stipulated time period and to produce cash/audit statements.

C7.3.2 *Vahan*

Vahan software has been implemented for the solution of vehicle registration related works. Software has provision of online scrutiny of the documents and booking of vehicle inspection date on the applicant's choice. Several types of MIS reports for the District administration and other agencies like police, insurance companies, etc. can be generated instantly.

C7.3.3 Certificates

Software for issuance of Certificates of Caste, Birth and Death, Haryana Residence has been implemented. The Kendra is issuing certificates instantly on submission of verified document and in case of non-verified documents, certificates are being issued within stipulated time period after getting the verification done from concerned officer.

C7.3.4 *Nakal*

Nakal of Jamabandies, Mutations and Khasra Girdawari are being issued from this counter within stipulated time frame. *Nakal* are being generated from the online computerized data available as well as from the manual records.

C7.3.5 Passport

Computerized file reference number is being issued to the Passport applicants and after scrutiny these are forwarded electronically to Regional Passport Office for issuance of passport.

C7.4 SELF SUSTAINABILITY

The entire project has been set up without any budgetary allocation from the Government. It is funded by District Red Cross Society on the lines of 'Haryana Registration Information System (HARIS)'. All running expenses including those of manpower are also being met by the District Red Cross Society. For

- Conducting Government's financial transactions such as payment of salaries, other payments like TA, LTC, contingent, rents, loans and advances, receipts of Government revenue, etc.
- Conducting of cash transactions through non-banking treasuries and banks
- Payment of pensions
- Pre-audit of bills and post-audit of vouchers
- Reconciliation of the department receipts and expenditure
- Safe custody of valuable articles such as election boxes, examination papers, department cash chest, treasuries trove, and the like
- Receipt, safe custody and sale of various kinds of stamps
- Payment of Government Securities such as Stock certificates, bearer bonds, promissory notes and income tax deductions
- Exercise treasury control over all departmental financial transactions with reference to Budget
- Maintenance of GPF account of last grade employees
- Maintenance of Employees Welfare Fund Account of employees
- Regulation of appointment to the public services through Act-2 of 1994
- Rendering of daily figures to Finance department
- Rendering of classified accounts to the AG
- Maintenance of deposit accounts including GO No. 43
- Maintenance of User charges Accounts
- Submission of PMES information to Government
- Enforcing accountability on DDOs and reporting to the Government as per GO No. 507
- Maintenance of employee census database

Accounting functions

Accounting functions include the following:

- Rendering financial advice to head of the departments
- Preparation of budget estimates of the respective departments
- Maintenance of departmental accounts in the respective departments
- Reconciliation of the receipts and expenditure of the department
- Processing of the pension cases of the respective departments
- Internal audit of the Department Unit Offices and follow-up action for settlement of the audit objections

Reports generated by treasuries and accounts department

- Classified Accounts to AG—STO/DTO Level
- GO No. 507—STO/DTO/RJD/DTA/PPO/APPO
- PMES—STO/DTO/RJD/DTA/PPO/APPO
- Daily figures—STO/DTO/DTA
- Receipt and Payment Reconciliation—DDO-HOA wise
- Pensions/strong room/stamps

- Budget vs. Expenditure—DDO wise and object headwise
- Non-Banking Sub treasuries
- Class IV G.P.F./E.W.F.
- Deposits Accounts and GO No. 43
- User Charges
- Employee Census
- Any other report required with available data

C11.2.2 System Overview

Budget authorization

The DTA, after exercising check on Distribution Statements furnished by HoD's with reference to Budget Releasing Orders (BRO) and Budget Estimates (BE) approved by the Assembly, issues authorizations to the DTO's concerned for Budget distribution made by the HoD.

Redistribution/Reauthorization

At any District, the subordinate controlling officer or the Head of the District, further distributes the Budget down the line to each Drawing officer and sends a copy of this budget distribution statement to the DTO. The DTO, after exercising proper cheques with reference to the Budget authorized by the DTA, issues re-authorization to the STO concerned.

The Treasury Officer permits the Drawing and Disbursing Officer (DDO) to draw the money to the extent of authorized Budget provision.

Honouring of claims against government

The DDO from each office presents their claims to the Treasury department for withdrawal of money for various purposes like "Payment of salaries, implementation of various schemes, meeting of day-to-day expenditure in the offices, etc." Based on the nature of the claim, different bill forms are used.

Treasuries honour these claims after scrutinizing the admissibility of the claim, financial powers of the drawing officer, sanction orders required, the genuineness of the claim, etc. besides availability of budget. After thorough examination of the claim presented by the DDOs, the Bill is passed or the Bill is returned when it is not in order duly quoting the authority for its return.

Acceptance of receipts

Receipts that accrue to the Exchequer are remitted through an instrument called challan. This challan form is used for all sorts of receipts like tax revenue or fees to be paid or remittance of a recovery.

Treasury checks the challan whether HoA is reflected on the challan with reference to the purpose of remittance and the Department on whose behalf the remittance is made. An entry is made in a register and serial number of the entry is noted down on the challan with date. The Treasury Officer in token of having

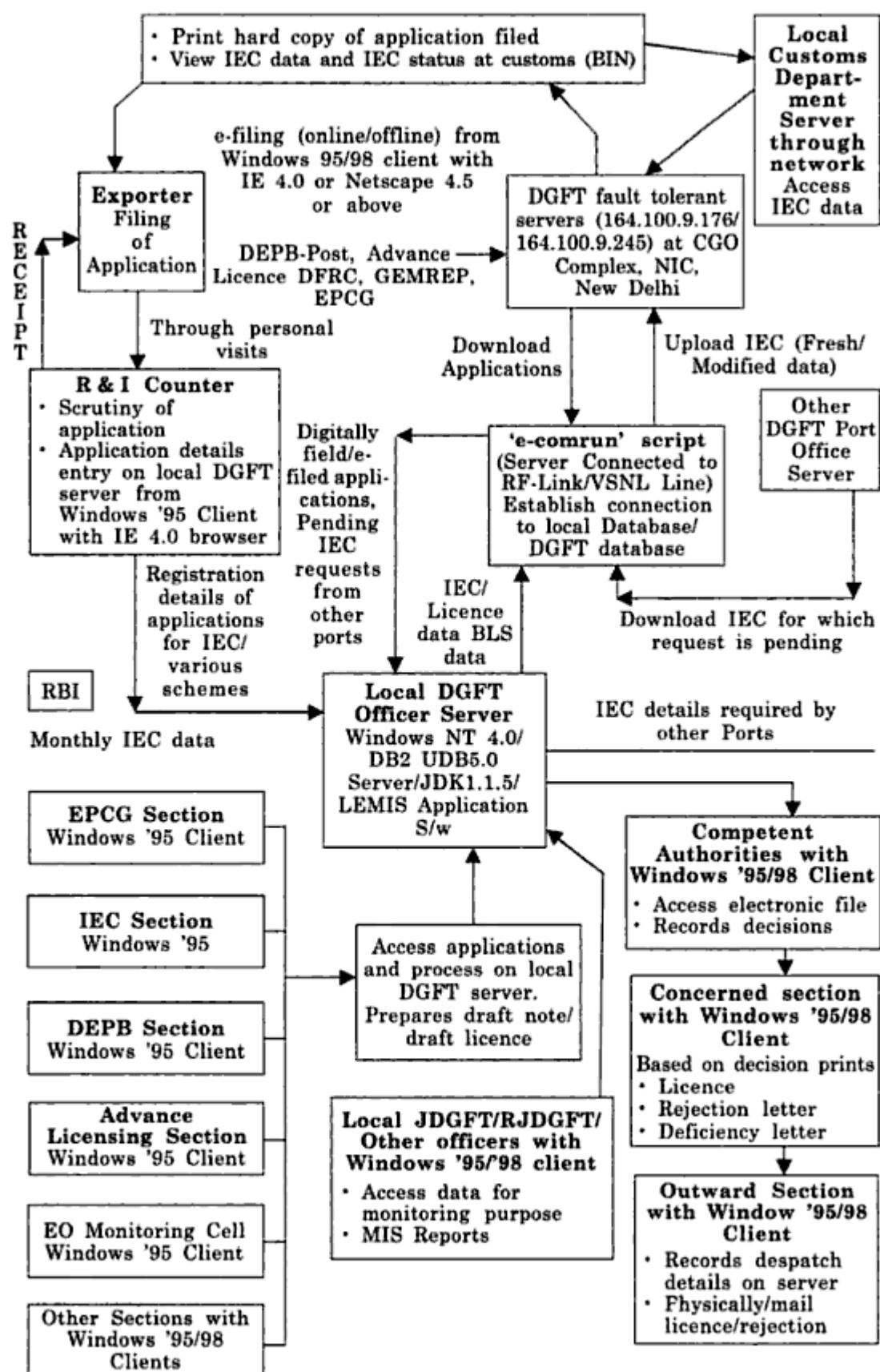


Figure C12.1 LEMIS Implementation Flow Diagram.

C12.8 POST COMPUTERIZATION SCENARIO

Improving administrative efficiency/services of trade by adopting IT in day-to-day functioning, is one of the objectives of computerization. It can be done by the following:

(i) E-filing of applications and E-commerce

Exporters now can file application electronically (online/offline modes). Within a day's time the application get processed in local DGFT office and licence is issued if details/documents are found correct. Only the concerned has to come to DGFT with necessary document and get his/her licence. Now DGFT is digital signature enabled; exporter can access and print licence at his end, and he needs to visit the office only once, that is, for submitting the necessary documents.

Apart from e-filing facility, DGFT has integrated the facility of digital signatures and electronic payment of licence fee in the online Electronic Data Interchange Module. DGFT has entered into an MOU with the State Bank of India, ICICI Bank, IDBI Bank, HDFC Bank and UTI Bank for Electronic Fund Transfer facility. To promote the facility of filing application using Digital Signatures, 50% reduction in licence fee is given with effect from 1st April 2004.

DGFT has tied up with Safecrypt Limited for issuance of Digital Certificate Services for filing of online applications for licences by EXIM community. Safecrypt has provided "SafeExim" Product package (which has already been integrated with LEMIS) for digitally signing and filing the online applications.

(ii) Electronic approval of licensing authority

LEMIS software has developed in such a way that the competent authority gets access to the application through his computer and approve the printing of licence by clicking approval button. Prior to this, dealing assistant has to generate office note, draft licence from computer, whereas now his efforts are reduced. This also amounts for automation in DGFT offices. Rejection/Deficiency letters are generated in the computer. History of application, like how many times it is updated/accessed and by whom are available in the database.

(iii) Importer-Exporter Code (IEC) and Binary Identification Number (BIN)

Every exporter has to obtain an Importer-Exporter Code Number in order to import/export any items under various licensing schemes. For fresh IEC or for any modifications of the firm details one has to apply local DGFT office. As soon as the details are fed/modifications are done, a script (java thread program) is invoked local DGFT computer and details are sent to DGFT server through

NICNET. DGFT Server, which in turn a BIN is allocated by computer. Customs will access such IECs information through DGFT web site and permit import/export of the items/goods. On the exporter's part also, one can view the IEC application details and know whether BIN number is generated or not and accordingly they can approach Customs Department. Here DGFT is rendering services to export/trade community/Customs Department Electronically/through Internet.

From 1st April 2004, DGFT is issuing a single IEC against a Single Income Tax PAN for all the category of applications. With this procedure, an exporter having different branches across the country can use same IEC number for his import/export activity, which was not the case earlier.

Now it is mandatory on the part of the exporter to file annual trade returns online.

(iv) Getting information for IECs registered other ports

Some times it may be required that local DGFT office has to download IEC details registered elsewhere any other port office in the country. This situation arises if branch office/factory is located in the jurisdiction of local DGFT and has applied for a licence for a particular scheme. Through Administration module of LEMIS request of IEC details are registered in the database and once 'e-commerce' script on the local is run, the required information is downloaded to the local server from other port database server.

(v) Licensing information at the disposal of DGFT

DGFT server, on a regular basis, gets details of licence issued at various port offices in the country through NICNET for further statistical analysis and dissemination.

Computerization also helps in providing information speedily to export community/trade. NIC-DGFT has hosted a web site on DGFT server and renders the following services:

- (E-filing of applications for DEPB-Post, Advance Licence, DFRC, GEMREP, EPCG) through online/offline modes
- Online help to e-filing (online/offline) for all existing schemes
- Viewing IEC details at exporter's end/customs end.
- To know IEC status/BIN number at Customs
- To get a digital certificate online
- Online filing of annual trade returns
- Uploading BRC data (for banks only)
- Online help for filing applications digitally/electronically
- Online check for Deletion Status in case of blacklisting (If a firm is blacklisted, the details are entered on local DGFT server, which in turn is transmitted to DGFT server at New Delhi on daily basis from all ports in the country. Any exporter can check whether his firm is blacklisted or not, through DEL status option of DGFT web site.)

CASE STUDY 14

Data Warehouse for the Ministry of Commerce

C14.1 INTRODUCTION

After the liberalization of the economy in 1991, foreign companies have shown keen interest in investing in India. Foreign investment approvals have gone up from US \$235 million in 1991 to US \$16281 million in 1995 and inflow of foreign direct investment has gone up from US \$155 million in 1991 to US \$4055 million in 1996 (till December). The investment boom has been injecting a high tide of competitiveness in Indian industries. In the wake of liberalization, many export-friendly schemes have been introduced. To give a further boost to exports, EXIM policies have been drastically liberalized and the procedural bottlenecks have been removed a great deal over the years. Amongst various export promotion schemes, export oriented unit (EOU) and export processing zone (EPZ) schemes occupy an important and distinct place in the country's export effort. The scheme offers an attractive package of incentives and facilities to investors.

The Ministry of Commerce has set up the following seven export processing zones at various locations as illustrated in Figure C14.1:

1. Kandla Free Trade Zone, Gandhidham
2. Santacruz Electronics Export Processing Zone, Bombay
3. Falta Export Processing Zone, Falta (West Bengal)
4. Madras Export Processing Zone, Chennai
5. Cochin Export Processing Zone, Cochin
6. Noida Export Processing Zone, Noida, (UP)
7. Visakhapatnam Export Processing Zone, Visakhapatnam

The export processing zones (EPZs) are set up to provide an internationally competitive duty-free environment for export production at low cost. These zones are responsible for the administration of free trade zones in setting up 100% export-oriented multiproduct industrial units. Each export processing

Table C16.1

<i>Centre</i>	<i>Address</i>
1. Banjara Hills	MCH Site, Road No. 7, Banjara Hills Tel: 23356648
2. Khairatabad	RTA Complex, Khairatabad Tel: 23310176
3. Marredpally	Opp St. Anns, HMWS & SB Site, Marredpally Tel: 27700753
4. Musheerabad	Beside Central Jail, HMWS & SB Site Tel: 27505236
5. Nallakunta	Near Shivam Road, HMWS & SB Site New Nallakunta Tel: 27402183
6. Ram Nagar	MCH site, Community Hall, Ram Nagar Ph: 27632217
7. Malakpet	Opp: Yashoda Hospital, Malakpet Tel: 24540195
8. Santhoshnagar	Adjacent to CRIDA, Santhoshnagar Tel: 24331635
9. Darulshifa	Salarjung Museum Road, Darulshifa Tel: 24526475
10. S.R. Nagar	HMWS & SB Site, S.R. Nagar X Roads Tel: 23701223
11. Sitaphalmandi	Near Water Tanki, Sitaphalmandi Tel: 27090361
12. Reti Bowli	APTRANSCO Site, Reti Bowli Tel: 23522134
13. Bahadurpura	Near Police Station, Bahadurpura Tel: 24471561
14. Vijaynagar Colony	MRO Complex, Vijaynagar Colony
15. Mint Compound	APTRANSCO Site, Mint Compound Tel: 23450073
16. Tirumalagiri	RTA Fast Track Premises, Tirumalagiri Tel: 27740172
17. Greenlands	Near Anand Nilayam Guest House, Greenlands Tel: 23400758
18. Sultan Bazar	APTRANSCO Site, Sultan Bazar Tel: 24606478
19. Habsiguda	APTRANSCO Site, Habsiguda Tel: 27206980
20. K.P.H.B.	Near Riytu Bazar, K.P.H.B. Colony Tel: 23055940

Continued

21. Vanasthalipuram	Behind Red Water Tank, Vanasthalipuram Tel: 24126733
22. Paradise	ERO Office, Paradise Tel: 27894736
23. Saroornagar	ERO Office, Saroornagar Tel: 24146957
24. Chikkadpally	MCH Complex, Chikkadpally Tel: 27608431
25. Motinagar	Beside Newton Public School, Motinagar Tel: 23841984
26. Kukatpally	Near Bus Stop, Kukatpally Tel: 23063849
27. Domalaguda	Near MCH Park, Domalaguda
28. Qutbullapur	Rythu Bazar, Qutbullapur
29. Ramanthapur	Near APSRTC, Ramanthapur
30. Sanatnagar	ERO office, Sanatnagar

Bank Branches

- | | |
|-----------------------|-----------------------------------|
| 1. Prakashnagar (AB) | Andhra Bank, Prakash Nagar Branch |
| 2. Jubilee Hills (AB) | Andhra Bank Jubilee Hills Branch |
| 3. Kushaiguda (SBH) | Near ECIL X Roads, Kushaiguda |
| 4. Malkajigiri (SBH) | Malkajigiri. |

C16.3.3 Transaction Details

- Number of transactions made from
25th August 2001 to March 2003 : 70.06 lakhs
Total value of transaction : Rs. 2333.33 crores

C16.3.4 Extension of Service Centres

Ranga Reddy district

Eighteen sites in eight municipalities in Ranga Reddy district have been identified. Out of 18 sites, six centres, namely, Habsiguda, KPHB, Vanasthalipuram, Saroornagar, Motinagar, and Kukatpally have been functioning since September 2002. The other centres are as follows:

- | | |
|--------------------------|------------------------|
| 1. Dilsukhnagar | 5. Qutubullapur-1 site |
| 2. Rajendranagar-2 sites | 6. Chanda Nagar |
| 3. Neredmet | 7. Kapra-2 sites |
| 4. Alwal | 8. Fateh Nagar |

C17.2.27 Weaker Sections Welfare

This module offers information services on:

- Atrocities reporting
- Welfare schemes
- Inventory and weaker artisans
- Marketing service
- Produce of weaker sections.



C17.2.28 Public Distribution

Information services available in this module are:

- FP shops
- Stock status
- Card holder information
- Problems report
- *Grihamitra*.



C17.2.29 Protection of Property

This module facilitates information services such as:

- Encroachment complex
- Status of encroachment
- Maintenance
- Tax collection bills.



Land Records module of e-Panchayat enables Land Records Maintenance by Village Secretary and rendering verification and certification services to the citizens. This also facilitates land revenue management, with appropriate reports to the revenue/gram panchayat officials and the citizens.

C17.2.30 Village Accounting System (VAS)

Following are the features of VAS:

- Cash transactions
- Bill receipts
- DDs and cheque receipts
- Automatic challan submission
- Treasury payments
- Classified accounts
- All kinds of taxes, fee collections (Online/Offline)
- Cheques/DDs reconciliation

- Reports (Daily collection report)
 - *Ersalnama*
 - *Chitta*
 - Monthly classifications
 - Challan receipts
 - Demand and collection reports.

C17.3 OTHER VILLAGE SERVICES (OVS)

C17.3.1 Births and Deaths Registrations

- Capturing of event details
- Name inclusion
- Non-availability certificates
- Registration of events
- Issuance of certificates
- Statistics.

C17.3.2 Property Tax and Vacant Land Tax

- Assessment as per monthly rental values
- Warrants and distraints
- Title transfers
- Demand collections
- Writeoff and temporary remissions
- Vacancy remissions
- Exemptions
- Special notices/endorsements/demand notices
- Registers/Field books/Defaulters.

C17.3.3 Dangerous and Offensive Trades

- Issue of licences
- Cancellations
- Renewal certificates
- Demand notices
- Fee collection.

C17.3.4 Water Tap Connections and Water Tax Collections

- Sanction of metered/Non-metered connections
- Disconnections/Reconnections

utilization, operational holdings and statewise agriculture output, i.e. area, product and yield of various crops.

Education

This is a comprehensive information portal on educational resources in India and abroad and is powered by up-to-date, reliable information. It is the best resource to the students, educational institutions, academic experts and all others in education field wherein a lot of information on courses, degrees, colleges, professional courses—statewise is made available. To know more about the career prospects in IT field after Intermediate, the separate module 'Education after 10 + 2' is the right choice to browse and know. Information about various vendor certified IT programs is also available.

Tourism

India being a paradise for tourists, GISTNIC has provided a lot of information about various tourist spots in India. One can categorically search based on a region (south/north/east/west) historic spots, religious spots etc. or make a statewise search. The data includes how to reach the spot, where to stay, important places to visit around it, climatic conditions and the like. Pictures were also included wherever available.

District profiles

This module gives various socioeconomic statistics and general information about a district of your choice.

Rural technologies

This module offers information on various technologies useful for rural development, employment generation, use of non-conventional and renewable energy.

Traditional sciences and technologies of India

This module provides information on traditional sciences and technologies of India in the field of medicine, metallurgy, architecture, agriculture and irrigation, textiles, and arts and crafts. A separate module is available to provide information about medicinal plants.

India's rich and varied heritage in Science and Technology has been universally recognized. Even upto 18th Century the technological culture in India has been identified as the forerunner of the Industrial Revolution in the West. In diverse fields of Science and Technology as Medicine, Metallurgy, Architecture, Textiles and theoretical sciences as Logic and Mathematics, India held a place of pride in the world before the Industrial Revolution. The survival of a large number of traditional methods and practices, especially systems such as Ayurveda, Yoga and Architecture, even after several thousand years, amply demonstrates the inherent strength and tenacity of traditional Indian sciences

and technologies. The westernized orientation of the present system of education and the cultural invasion of the West in all channels and directions has resulted in almost a final detachment of the common man, especially the younger generation, from the Traditional Sciences and Technologies of India. This has resulted in a great loss, not only to the Indian scientific spirit and technological capabilities in the country, but also in affecting the developmental planning and implementation.

However, an appropriate role the Traditional Sciences and Technologies can play in rural development has been recognized of late, and various efforts towards promoting the Traditional Sciences and Technologies have now begun. In this direction NIC has taken up the task of creating and disseminating a computer based database on Traditional Sciences and Technologies of India under the GISTNIC. This database has been classified into about 10 major subject areas and 100 minor subject areas.

GISTNIC-STD code enquiry service

This allows the user to search for an STD code. By giving the first 3 characters (or the Exact name) of a place or by giving a STD code you can know the place name. You can even search by complete number (complete number as per the telephone bill and search). This makes the user to know the code of the place for which the call was made. Links to various DOT sites is also made available from which one can search for a phone number or name anywhere in India.

Companies and industries directory

This provides productwise company profiles and addresses, while exports statistics include the trends and scenario in Indian Exports.

All India primary census abstract 1991

The abstract of 1991 census is also made available keeping in view the importance of the census statistics. All India, statewise, districtwise, mandalwise, talukawise, townwise and villagewise details of the primary census is made available under this module.

Engineering and MBBS admissions

These are very crucial moments for the aspiring engineering and medical student community. Realizing the need and importance of this, GISTNIC project added a separate portal service. Once the EAMCET admissions start in AP student community will be anxious to know the seat vacancy position in various courses and colleges—reservation categorywise. All such vacancy position details are well updated on the EAMCET site under GISTNIC umbrella. Data will be updated right from the admission centre. Every year, lakhs of people throng the GISTNIC site during admission time to know the latest status of admission. Same is the case with MBBS admissions. While

Architecture (FEA)—a comprehensive view of what government organization does, how does it do and how IT supports it. The FEA is a business focussed framework that provides the Office of Management and Budget (OMB), which is the agency responsible for FEA and the Federal Agencies with a methodology to monitor, analyze and control federal investments in IT. The FEA enables horizontal (Cross Federal) and vertical (Federal, State and Local governments) collaboration and communications with respect to IT investments.

C19.2.2 Implementation of President's Management Agenda

The e-governance initiative will enable implementation of various parts of Presidents Management Agenda (PMA):

Budget and performance integration, strategic management of human resources, competitive procurement of services and bidding, and improved financial performance. E-payroll, e-training, (e-learning), e-clearance, and e-procurement.

C19.2.3 Performance Measurement in E-Governance Implementations

A quarterly scoreboard is used to monitor the performance of various federal agencies engaged in e-governance implementation in their respective departmental activities. Standard requirements require the federal agencies to fulfill at least three of the following criteria:

1. Integration of citizen's one-stop service delivery through *Firstgov.gov* and also call centres or service centres
2. Minimization of burden on business by re-using data previously collected or using XML to receive transmissions and data
3. Inter governmental deployment of e-grants or geospatial outsourcing
4. Achieving productivity, improvements by implementing Customers Relationship Management (CRM), Supply Chain Management (SCM), Knowledge Management (KM), and Enterprise Resource Planning (ERP).

A Federal Agency will be given 'green' score if it meets all the above standards for success. 'Yellow' score will be given if the agency had partially achieved this and 'red' will be given if there exist significant gaps between the goals set and ground reality in the agency.

A number of agencies have shown improvement from red to yellow such as Department of Education, Department of Energy, and Department of External Affairs. NSF (National Science Foundation) upgraded itself from yellow to green.

Subsequently, 'green' was given to a number of agencies such as Department of Housing and Urban Development, Development of Interior,

Department of Justice, Department of Agriculture, Department of Labour, Department of Transportation, Environmental Protection Agency, NASA, NSF, and Social Security Administration.

C19.2.4 E-Government Projects, Portfolios or Categories

The e-government projects are organized into portfolios or categories as defined by the customers requirements served. The following categories are described in terms of their goals:

Government to Citizens (G2C)

The objective of (G2C) category of services is to provide one stop online access to governmental information and services to individual citizens. In this the citizens are expected to access and obtain any information in seconds as against days and weeks. For examples *Govbenefits.gov* portal provides instantaneous access to citizens regarding information about all government programmes and services through a single web site within three clicks.

Government to Business (G2B)

The objective of (G2B) portfolio or category of services is to provide one stop access to business community on information pertaining to all government procedures, and rules and regulations pertaining to them. The previous practice was making businesses submit multiple forms of data multiple times to different government agencies. The objective is to provide a single window interface with the government, including all departments.

The government should re-use the data submitted only once for different multiple purposes. This regulates appropriate interpretation of the same data entered in various formats. The deployment of a more effective technology will help streamline the myriad reporting requirements, as well as facilitate a more efficient means for business to interact with the government. Citizens or Businesses can easily find rules and regulations at *www.regulation.gov* even without the help of an attorney or a lawyer.

Government to Government (G2G)

The objective of this portfolio is to enable different wings of the federal government or the State government interact effectively with each other. To achieve this, the objective of the Federal Government is to make it easier for states to meet the reporting requests while prompting the use of performance monitoring. This will definitely result in substantial savings for the State governments. The data delivery will be more accurate and time saving. An example in this direction will be Home Land Security Initiative, a security portal that will improve disaster management process, by simplifying and unifying the interaction between various governmental agencies as Federal, State and Local administration as well as Local Safety and Security Organizations and personnel.

Table C19.3 underlines the feature of e-authentication.

Table C19.3

E-Authentication

- Reduce the number of credentials by customer segment needed to interact with the Federal government

E-government accomplishment in USA

The following e-governance projects throw light on the e-governance initiatives in USA:

- (i) ***FirstGov.gov.*** FirstGov.gov is an American citizens' gateway to about 200 million pages of Information distributed in more than 22,000 Federal and State government web sites. Named as one of the "Top 50 most incredibly useful web sites" in the entire world by Yahoo in 2002, this service is designed to provide government service within *three clicks*. The new "three clicks" strategy has drawn dramatically large number of hits upto 30 millions.
- (ii) ***Volunteer.gov.*** The site enables American citizens interested in volunteering to work for more than 100,000 openings available in government hospitals and other Federal organizations.
- (iii) ***Recreation.gov.*** The site provides citizens online access to more than the total 22,000 National Parks and Recreation areas of US.
- (iv) ***Gov.Benefits.gov.*** This site provides access to more than 400 government programmes of benefits. It receives about 500,000 visitors per month and is listed as a 'Hot site'.
- (v) ***IRS free filing.*** Over 80 million Americans can file their taxes online for free.
- (vi) ***Integrated Acquisition Environment (IAE).*** This site provides information on past performance of contractors and also Federal technical data on acquisition material.
- (vii) ***Business.Law.gov.*** This site is designed to provide small businesses quick access to legal and regulation information tools of compliance, the ability to perform online transactions and business related information of the government.
- (viii) ***Regulation.gov.*** The site provides data on all government regulations. This has received 2.6 millions hits in January 2003.
- (ix) ***GoLearn.gov.*** This is the world's most visited e-training initiative site with more than 60 millions hits, with information over 2000 e-training courses, e-books and carrier development resources. This site has more than 45,000 registered users who can receive training at negligible costs.

C20.2.1 The Old System of Administration

The old system of administration was very inefficient and was distributed. All the 12 departments acted independent of each other. Each had its private database and Information Systems on its own independent computer systems. There was no integrated workflow management across the departments.

The users of old systems found it extremely difficult to interact with the administration. For example, for obtaining a licence to open a company in the park, the company had to apply to eight different departments and make at least three visits to each department.

First, they had to apply to the first department with a bunch of attachments. If found acceptable, after about 10 days they will be informed that their application is accepted. Then the story repeats for the second department also. The total time taken (for a successful case) will be around three months.

Evidently three months delay in obtaining licence with substantial harassment was quite painful and a great avoidable burden on the companies concerned. This was adversely affecting the growth of Chinese Economy, as a whole. Companies in the Park generated a total of \$12 billion as revenue in the year 2000 and \$200 million foreign investments. Clearly, the contribution of the park was highly significant for Chinese Economy and therefore the government took a decision to streamline the situation.

C20.2.2 The New System

The administration of the park formed a strategic partnership with a local private application software company, Beijing Beauty Bearl limited which started the system study and completed the software development by the end of the year 2000.

In the new system the central database and web site allow data sharing and workflow integration among all the departments. Thus the administration functions are a single integrated body showing only one face to the public. The system includes five functions, all accessed from the web site homepage of the web site.

The following are the five functions of the system.

(i) E-application

This is a web based software that provides the applicants with all forms and documents required, along with rules and regulations, and also procedures for setting up a company in the Park. After login with the user ID and password the users can fill up the files directly on the web or download the forms, fill them up on the computer, and upload them into the web. The form will depend upon the nature of the company. Almost instantaneously, the ZSP Administration Committee receives the applications and begins approval procedures online. Two or three layers of approval are processed by the Administration. In each layer the Administration sends its comments/feedback/objectives to

